Practice workbook · Trig equations that factorise

Solving an equation by factorising out a common trig factor (or using a product form) and solving each factor, instead of dividing and losing solutions.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-26

1. Question 8 · part b · 4 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part b only.

8 (a) Solve, to the nearest integer, the equation

$$\sin(2x - 75)° = -0.515 \quad \text{for } 0 \leqslant x < 180$$

(3)

(b) Giving your solutions to one decimal place, where appropriate, solve the equation

$$2\tan y° + 5\sin y° = 0 \quad \text{for } 0 \leqslant y \leqslant 180$$

(4)

(c) Explain mathematically why there are no values of θ that satisfy the equation

$$3\cos^2\theta° - 3\sin^2\theta° + \sin\theta° + 12 = 0$$

(4)

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1. Question 8 · part b · 4 marks · Working and answer

Working and answer

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2. Question 1 · part whole · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part whole only.

1 Solve the equation

$$\sqrt{3}\tan A = 2\sin A \quad \text{for} \quad 0^\circ \leqslant A \leqslant 360^\circ$$

(5)

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(Total for Question 1 is 5 marks)

2. Question 1 · part whole · 5 marks · Working and answer

Working and answer

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3. Question 7 · part b · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

7 (a) Solve the equation

$$\cos(3x - 15)° = \frac{\sqrt{3}}{2} \quad \text{for} \quad 0 \leqslant x < 180$$

(4)

(b) Solve, giving your solutions to one decimal place where appropriate,

$$3\tan y° + 4\sin y° = 0 \quad \text{for} \quad -180 \leqslant y < 180$$

(4)

(c) Solve, giving your solutions to one decimal place where appropriate,

$$\cos\theta° = 3\sin^2\theta° - 1 \quad \text{for} \quad -180 \leqslant \theta < 180$$

(4)

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3. Question 7 · part b · 4 marks · Working and answer

Working and answer

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4. Question 10 · part c · 4 marks

2023 JAN · Paper 02

Further practice · Attempt part c only.

10 Using formulae given on page 2

(a) show that

(i) $\sin 2\theta = 2\sin\theta\cos\theta$

(ii) $\cos 2\theta = 2\cos^2\theta - 1$

(5)

Given that $\theta \neq (90° + 180°n)$ where $n \in \mathbb{Z}$

(b) use the results from part (a) to show that $\sin 2\theta - \tan\theta$ can be written as $\tan\theta\cos 2\theta$

(4)

(c) Solve for $0 < x < 360$

$$\sin 2x° - \tan x° = 0$$

(4)

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4. Question 10 · part c · 4 marks

2023 JAN · Paper 02

Further practice · Attempt part c only.

Question 10 continued

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4. Question 10 · part c · 4 marks

2023 JAN · Paper 02

Further practice · Attempt part c only.

Question 10 continued

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4. Question 10 · part c · 4 marks

2023 JAN · Paper 02

Further practice · Attempt part c only.

Question 10 continued

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(Total for Question 10 is 13 marks)

4. Question 10 · part c · 4 marks · Working and answer

Working and answer

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5. Question 10 · part c · 4 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

10 (a) Show that

$$\cos(A + B) + \cos(A - B) = 2\cos A \cos B$$

(2)

(b) Hence show that

$$\cos P + \cos Q = 2\cos\frac{P + Q}{2}\cos\frac{P - Q}{2}$$

(3)

(c) Solve, for $0 \leqslant \theta \leqslant \frac{\pi}{2}$, the equation

$$\cos 5\theta + \cos 7\theta = 0$$

Give each solution in terms of π

(4)

(d) Show that

$$\cos 8x + 2\cos 6x + \cos 4x = 4\cos 6x \cos^2 x$$

(3)

(e) Use calculus to find the exact value of

$$\int_0^{\frac{\pi}{3}} \cos 6x \cos^2 x \, dx$$

(4)

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5. Question 10 · part c · 4 marks · Working and answer

Working and answer

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6. Question 8 · part c · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

8 (a) Write down the value of k such that $\sin 2A = k \sin A \cos A$

(1)

$$g(A) = 2 + 3\cos A - \sin A - 3\sin 2A - 2\cos^2 A$$

Given that $g(A)$ can be written in the form $(p\cos A - \sin A)(q - r\sin A)$ where p, q and r are integers,

(b) find the value of p, the value of q and the value of r.

(3)

(c) Hence solve, in radians to 3 significant figures where appropriate, the equation

$$g(2\theta) = 0 \quad \text{for} \quad 0 \leqslant \theta < \pi$$

(6)

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6. Question 8 · part c · 6 marks

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Further practice · Attempt part c only.

Question 8 continued

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Turn over ▶

6. Question 8 · part c · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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6. Question 8 · part c · 6 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 8 continued

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(Total for Question 8 is 10 marks)

6. Question 8 · part c · 6 marks · Working and answer

Working and answer

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7. Question 10 · part c · 7 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

10 Using formulae from page 2

(a) show that $\cos(A - B) - \cos(A + B) = 2\sin A \sin B$

(2)

(b) Hence show that $\cos 5\theta - \cos 9\theta = 2\sin 7\theta \sin 2\theta$

(1)

(c) Solve the equation

$$\cos 5\theta - \cos 9\theta = \sqrt{3}\sin 7\theta \quad \text{for} \quad 0 < \theta \leqslant \frac{1}{3}\pi$$

Give your solutions in terms of π

(7)

(d) Using calculus and showing your working, evaluate

$$\int_0^{\frac{\pi}{7}} 8\sin 7x \cos 2x \tan 2x \, dx$$

Give your answer to 3 decimal places.

(6)

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7. Question 10 · part c · 7 marks · Working and answer

Working and answer

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